

Craniometadiaphyseal Osteosclerosis with Hip Dysplasia (CMDOH): A Comprehensive Disease Characterization

Disease: Craniometadiaphyseal Osteosclerosis with Hip Dysplasia (CMDOH) **MONDO ID:** MONDO:0957832 · **OMIM:** #620558 · **Gene:** *AXIN1* (OMIM 603816; HGNC:903; chr16p13.3)
*Category:** Mendelian (autosomal recessive)

Summary

Craniometadiaphyseal osteosclerosis with hip dysplasia (CMDOH) is an ultra-rare, autosomal recessive **sclerosing skeletal dysplasia** caused by biallelic (homozygous, in the reported consanguineous families) **C-terminally truncating variants in *AXIN1***. *AXIN1* is the scaffold protein of the β -catenin destruction complex, the negative regulator of canonical Wnt signaling. The disease-causing variants truncate the **C-terminal DIX domain**, which mediates *AXIN1* self-polymerization, thereby lowering *AXIN1* protein levels and impairing destruction-complex assembly. The net functional consequence is **enhanced basal canonical Wnt/ β -catenin signaling** — a "Wnt-gain" state — that shifts the balance of bone remodeling toward increased osteoblastic bone formation and reduced osteoclastic resorption, producing generalized osteosclerosis and hyperostosis.

The condition was defined in 2023 by Terhal et al. ([PMID: 37582359](#)), who described **seven individuals from four families** carrying **three distinct homozygous truncating *AXIN1* variants**. The core phenotype comprises congenital/progressive **macrocephaly**, **cranial hyperostosis** (calvarial and skull-base sclerosis), and **vertebral endplate sclerosis** ("sandwich vertebrae"), together with frequent **hip dysplasia**, **heart malformations**, **variable developmental delay**, **hematological anomalies**, metaphyseal flaring, and dysmorphic facies. Bone biopsy demonstrated increased osteoblast activity and reduced osteoclast function at the growth-plate resorption zone, consistent with the Wnt-gain mechanism.

Mechanistically, CMDOH sits at the "hypomorphic" end of an *AXIN1* dysfunction spectrum: germline biallelic DIX-domain truncations are partial loss-of-function (protein is reduced and polymerization is impaired but some Wnt-inhibitory function is retained), yielding a viable skeletal phenotype, whereas **complete somatic loss of AXIN1** drives cancer (notably hepatocellular carcinoma) through full canonical Wnt activation. Crucially, the same study showed that a **tankyrase inhibitor (XAV939)** — which stabilizes AXIN1/AXIN2 — attenuates the Wnt overactivity in patient-derived and genome-edited cells, nominating a mechanistically rational (though not yet clinically tested) targeted therapy. Mouse *Axin1* models corroborate the β -catenin-dependent, osteoblast-lineage mechanism. No disease-specific therapy currently exists; management is supportive and multidisciplinary.

Section 1 — Disease Information

Overview. CMDOH is a rare Mendelian sclerosing bone dysplasia combining features of two classic radiographic categories: osteopetrosis-like changes (sandwich vertebrae, metaphyseal flaring) and endosteal-hyperostosis-like changes (calvarial and cortical thickening). It is characterized by macrocephaly, cranial hyperostosis, vertebral endplate sclerosis, hip dysplasia, and additional extraskeletal involvement (cardiac, developmental, hematological). "Sclerosing skeletal dysplasias result from an imbalance between bone formation and resorption" ([PMID: 37582359](#)) — CMDOH exemplifies this, with the imbalance tilted toward net bone accrual.

Key identifiers.

Resource	Identifier
MONDO	MONDO:0957832
OMIM (phenotype)	#620558 (Craniometadiaphyseal osteosclerosis with hip dysplasia)
Gene	<i>AXIN1</i> , OMIM *603816, HGNC:903, chromosome 16p13.3
Orphanet	No dedicated ORPHA code clearly assigned at time of investigation (ultra-rare, delineated 2023)
ICD-10 / ICD-11	No specific code; maps under sclerosing/osteosclerotic bone dysplasia (e.g., ICD-10 Q78.-, "Other osteochondrodysplasias")
MeSH	No specific descriptor; related: "Osteosclerosis", "Osteochondrodysplasias"

Synonyms / alternative names. Craniometadiaphyseal osteosclerosis with hip dysplasia; CMDOH; *AXIN1*-related sclerosing skeletal dysplasia. The name captures the anatomic distribution (**cranio-** = skull; **meta-/diaphyseal** = metaphysis/diaphysis of long bones) plus the associated hip dysplasia.

Information source. Knowledge is derived from an **aggregated case series** — deep clinical, radiographic, and functional characterization of individual patients (7 individuals / 4 families) reported in a single landmark paper, plus mechanistic corroboration from cellular models and model organisms. It is **not** derived from large EHR/registry datasets, reflecting the disorder's extreme rarity.

Section 2 — Etiology

Primary cause (genetic). CMDOH is a **monogenic autosomal recessive** disorder caused by **biallelic C-terminally truncating variants in *AXIN1***. In the reported families, affected individuals were homozygous, consistent with parental consanguinity. There is no known environmental, infectious, or acquired cause.

Genetic risk factors. The disease is fully genetically determined by the *AXIN1* genotype. **Consanguinity** is the principal contextual risk factor, increasing the probability of homozygosity for a rare recessive allele. No modifier loci have been formally mapped; the paralog **AXIN2** is a plausible biological modifier because it can partially compensate for *AXIN1* (see Mechanism).

Environmental risk factors. None identified. Sex does not appear to influence occurrence (recessive Mendelian trait). Family history / consanguinity is the relevant contextual factor.

Protective factors. No genetic or environmental protective factors are established. Biologically, retained partial *AXIN1* function and *AXIN2* compensation are inferred to make the phenotype viable (compared with the lethality of complete *Axin1* loss in mice), but this is a feature of the hypomorphic allele class rather than a modifiable protective factor.

Gene–environment interactions. None demonstrated. The disorder is essentially environment-independent.

Section 3 — Phenotypes

The phenotype spectrum below is drawn from the 7 patients / 4 families in Terhal et al. 2023 ([PMID: 37582359](#)). Frequencies are qualitative given the tiny cohort.

Phenotype	Type	Onset	Severity / progression	Frequency	Suggested HPO term
Macrocephaly (+2.2 to +6.1 SD)	Physical/clinical sign	Congenital, progressive	Moderate–severe, progressive	Core / very frequent	HP:0000256
Cranial hyperostosis (calvarial + skull-base sclerosis)	Imaging/clinical sign	From first year, progressive	Progressive	Core / very frequent	HP:0004437
Vertebral endplate sclerosis ("sandwich vertebrae")	Imaging sign	Childhood	Progressive	Core / frequent	HP:0004618
Metaphyseal flaring / widening	Imaging sign	Childhood	Progressive	Frequent	HP:0003015
Hip dysplasia	Clinical/imaging sign	Congenital/childhood	Variable	Frequent	HP:0001385
Heart malformations	Structural anomaly	Congenital	Variable	Frequent	HP:0001627
Developmental delay	Neurobehavioral	Childhood	Variable/mild-moderate	Variable	HP:0001263
Hematological anomalies	Laboratory abnormality	Variable	Variable	Frequent	HP:0001871
Hypertelorism	Dysmorphic sign	Congenital	Stable	Frequent	HP:0000316
Depressed/low nasal bridge	Dysmorphic sign	Congenital	Stable	Frequent	HP:0005280
Short stature (subset)	Growth	Childhood	Variable	Subset	HP:0004322

Onset and severity. Reported ages span **3 months to 15.5 years**. Macrocephaly is congenital and progressive; cranial hyperostosis emerges and progresses from the **first year of life**. Severity is variable across families.

Quality-of-life impact. No formal QoL instruments (EQ-5D, SF-36, PROMIS) have been applied. Anticipated impacts, by analogy with other cranial hyperostoses, include potential cranial-nerve compression (from skull-base sclerosis), orthopedic morbidity from hip dysplasia (pain, mobility limitation), and developmental/educational impact where developmental delay is present. These are inferred rather than measured.

Section 4 — Genetic / Molecular Information

Causal gene. *AXIN1* (axis inhibitor 1) — OMIM 603816, HGNC:903, chromosome 16p13.3*. *AXIN1* is the central scaffold of the β -catenin destruction complex (with APC, GSK3 β , CK1), which phosphorylates β -catenin to target it for degradation.

Pathogenic variants (Terhal et al. 2023). Three distinct **C-terminal truncating** variants across four families:

Family	Variant	Type	Consequence
Family 1 (two sisters)	1-bp duplication (frameshift, e.g., p.Asp796Glufs*6-type)	Frameshift	Premature truncation, DIX domain lost
Families 2 & 4	p.Arg723* (R723X) — recurrent nonsense	Nonsense	Truncation removing DIX domain
Family 3	p.Arg805* (R805X)	Nonsense	C-terminal truncation, DIX domain disrupted

All variants disrupt the **C-terminal DIX domain**, which mediates *AXIN1* homo-polymerization and destruction-complex assembly.

Variant classification. Per ACMG/AMP criteria these are **pathogenic** (loss-of-function truncating variants in a gene with an established gene–disease relationship, segregating with disease in multiple consanguineous families, supported by functional evidence).

Allele frequency. These are private/ultra-rare variants; not present at appreciable frequency in gnomAD. The recurrent p.Arg723* appearing in two families likely reflects an arginine CpG mutational hotspot rather than a broad founder effect.

Somatic vs germline. In CMDOH the variants are **germline**. By contrast, *AXIN1* is recurrently altered **somatically** in cancer (see F003).

Functional consequence. Hypomorphic loss of function. The truncations reduce *AXIN1* protein levels and impair DIX-domain-mediated polymerization, but overexpression studies showed partially **retained** Wnt-inhibitory function — explaining why the phenotype is a viable skeletal dysplasia rather than the embryonic lethality seen with complete *Axin1* loss. Net pathway effect: **enhanced basal canonical Wnt/ β -catenin signaling** (gain of Wnt output via loss of a negative regulator).

Modifier genes. *AXIN2* (the paralog) is the strongest candidate modifier: XAV939 rescue of *AXIN1*-knockout cells implicates *AXIN2* stabilization as a compensatory route (see F006).

Epigenetic / chromosomal. No disease-specific epigenetic signature or large-scale chromosomal abnormality has been reported for CMDOH.

Section 5 — Environmental Information

No environmental, lifestyle, or infectious contributors are implicated. CMDOH is a purely genetic, autosomal recessive Mendelian disorder. **Not applicable:** toxins/radiation/pollution; smoking/diet/exercise/alcohol; infectious agents.

Section 6 — Mechanism / Pathophysiology

Causal chain (initiating lesion → clinical manifestation)

1. **Biallelic C-terminal truncating variant in *AXIN1*** (e.g., p.Arg723, p.Arg805, or a frameshift) → **removes/disrupts the DIX domain**.
2. Loss of the DIX domain **impairs *AXIN1* self-polymerization** and **lowers *AXIN1* protein levels** → *results in* a weakened, less abundant scaffold.

3. Reduced functional AXIN1 → **destabilizes the β -catenin destruction complex** (AXIN1–APC–GSK3 β –CK1) → *leads to* reduced phosphorylation/degradation of β -catenin.
4. → **Cytoplasmic β -catenin accumulates and translocates to the nucleus** → *results in enhanced basal canonical Wnt/ β -catenin transcriptional output* (demonstrated directly in patient and genome-edited cells).
5. In the **osteoblast lineage**, elevated β -catenin → *increases OPG (osteoprotegerin)* and the **OPG:RANKL ratio** (inferred from mouse *Axin1^{ΔOsx}* model) → *leads to suppressed osteoclastogenesis* (fewer TRAP+/MMP9+/cathepsin-K+ osteoclasts).
6. Branch A — **increased osteoblastic bone formation**; Branch B — **decreased osteoclastic bone resorption**. Both branches converge to → **net increase in bone mass / osteosclerosis and hyperostosis**.
7. Anatomically distributed → **cranial hyperostosis + macrocephaly, vertebral endplate ("sandwich") sclerosis, metaphyseal flaring**; with associated developmental effects producing **hip dysplasia, cardiac malformation, developmental delay, hematological anomalies** (the extraskeletal links are consistent with the broad developmental role of Wnt signaling but are *inferred*, not individually demonstrated).
8. **Therapeutic branch: Tankyrase inhibition (XAV939)** raises AXIN1/AXIN2 levels → *restores* destruction-complex activity → *attenuates* Wnt overactivity in mutant cells (in vitro proof of concept).

Steps 1–4 and step 8 are **experimentally demonstrated in the human/cell system**; step 5 is **inferred from mouse osteoblast-lineage models**; step 7's extraskeletal links are **inferred** from Wnt biology.

Detail by category

- **Molecular pathway:** Canonical **Wnt/ β -catenin** signaling (KEGG hsa04310; Reactome "Degradation of beta-catenin by the destruction complex," R-HSA-195253). AXIN1 is the rate-limiting scaffold of the destruction complex.
- **Cellular processes:** Osteoblast differentiation/activity (increased); osteoclastogenesis (suppressed via OPG/RANKL). GO:0060070 (canonical Wnt signaling pathway), GO:0001649 (osteoblast differentiation), GO:0030316 (osteoclast differentiation), GO:0045597/0045668 (regulation of osteoblast/osteoclast differentiation).
- **Protein dysfunction:** DIX-domain truncation → **loss of polymerization**, reduced protein stability → hypomorphic LOF of the destruction-complex scaffold.

- **Biochemical abnormality:** Failure to phosphorylate/degrade β -catenin; downstream Wnt target activation (e.g., c-Myc, CCND1 seen in AXIN1-loss cancer models, supporting pathway direction).
- **Tissue remodeling mechanism:** Dysregulated bone remodeling with osteoblast–osteoclast imbalance at the growth-plate resorption zone; coarse trabeculae on biopsy.
- **Cell types (CL):** osteoblast (CL:0000062), osteoclast (CL:0000092), osteoprogenitor/mesenchymal stromal cell (CL:0000134), osteocyte (CL:0000137).

Molecular profiling

- Direct **functional readouts** (Wnt reporter activity) in patient-derived and CRISPR-edited cells: *enhanced basal canonical Wnt activity* (PMID: 37582359).
- Mouse osteoblast-lineage models: increased β -catenin, increased OPG:RANKL, decreased osteoclast markers.
- No dedicated patient transcriptomic/proteomic/metabolomic/single-cell datasets are available for CMDOH.

Section 7 — Anatomical Structures Affected

Organ / system level (primary). Skeletal system — with a distinctive craniospinal and metadiaphyseal distribution: - **Skull / calvarium & skull base** (UBERON:0000209 cranial skeletal system; UBERON:0004339 calvaria) — hyperostosis, macrocephaly. - **Vertebral column** (UBERON:0001130) — endplate sclerosis ("sandwich vertebrae"). - **Long bones — metaphysis/diaphysis** (UBERON:0002515 metaphysis; UBERON:0004770 diaphysis) — flaring, sclerosis. - **Hip joint / pelvis** (UBERON:0001464 hip; UBERON:0001474 bone of the pelvis) — hip dysplasia.

Secondary organ involvement. Cardiovascular (heart malformations, UBERON:0000948); hematopoietic system (hematological anomalies, UBERON:0002390); central nervous system (developmental delay; potential cranial-nerve compromise from skull-base sclerosis).

Tissue / cell level. Bone (connective) tissue; growth-plate cartilage/resorption zone. Cell populations: **osteoblasts** (increased activity), **osteoclasts** (reduced number/function), osteoprogenitors.

Subcellular level. β -catenin destruction complex is **cytoplasmic**; downstream signaling acts in the **nucleus** (β -catenin/TCF transcription). GO cellular components: GO:0030877 (beta-catenin destruction complex), GO:0005737 (cytoplasm), GO:0005634 (nucleus).

Localization / lateralization. Skeletal involvement is **generalized and bilateral/symmetric** (calvarium, spine, long bones); hip dysplasia may be uni- or bilateral.

Section 8 — Temporal Development

- **Onset: Congenital** (macrocephaly present at/near birth); cranial hyperostosis develops and progresses from the **first year of life**. Onset pattern is **chronic and insidious/progressive**.
- **Progression:** Radiographic sclerosis is **progressive** through childhood (cases documented from 3 months to 15.5 years). Progression rate appears slow-to-moderate and **variable** between families.
- **Disease course: Chronic, lifelong.** No episodic or relapsing-remitting pattern.
- **Remission:** None spontaneous; no treatment-induced remission described (no disease-modifying therapy in clinical use).
- **Critical periods:** The **first years of life** (active bone modeling) represent the window in which hyperostosis emerges — mechanistically the most rational period for any future targeted (Wnt-normalizing) intervention.

Section 9 — Inheritance and Population

- **Epidemiology: Ultra-rare.** Prevalence/incidence are unknown and unquantified; total literature comprises 7 individuals from 4 families. No registry-based estimate exists.
- **Inheritance: Autosomal recessive.** Affected individuals were **homozygous**; parents are obligate heterozygous carriers.
- **Penetrance:** Appears **complete** in biallelic individuals (all reported homozygotes affected), though the sample is too small for precise estimation.
- **Expressivity: Variable** — e.g., variable developmental delay, cardiac involvement, and stature across families/individuals.
- **Genetic anticipation:** Not applicable (not a repeat-expansion disorder).

- **Germline mosaicism:** Not reported.
 - **Founder effect:** No broad founder effect established; the recurrent p.Arg723* likely reflects a CpG mutational hotspot rather than a shared haplotype.
 - **Consanguinity: Central** — homozygosity arose in consanguineous families.
 - **Carrier frequency:** Unknown; expected to be very low given ultra-rarity.
 - **Population demographics:** Reported families are consistent with consanguineous backgrounds; no established sex bias (recessive). Distribution is sporadic, not endemic.
-

Section 10 — Diagnostics

Imaging (cornerstone of diagnosis). Skeletal survey / radiographs demonstrating: - Calvarial and skull-base **hyperostosis/sclerosis**; macrocephaly. - **Vertebral endplate sclerosis** ("sandwich vertebrae"). - **Metaphyseal flaring** and long-bone sclerosis. - Hip dysplasia on pelvic imaging. CT is useful to characterize skull-base sclerosis and potential foraminal narrowing.

Laboratory / biomarkers. No specific diagnostic biomarker. Bone turnover markers are not established as diagnostic. Hematological anomalies warrant a CBC and hematologic workup. Bone biopsy (research setting) showed **increased osteoblast and reduced osteoclast function** with coarse trabeculae.

Genetic testing (confirmatory). The diagnosis is molecularly confirmed by identifying **biallelic C-terminal truncating AXIN1 variants**. - WES/WGS is the practical first-line approach given phenotypic overlap among sclerosing dysplasias. - **Targeted AXIN1 sequencing** or inclusion in a **sclerosing bone dysplasia / skeletal dysplasia gene panel** is appropriate once suspected. - CMA/karyotype/FISH/mtDNA/repeat testing are **not** indicated (single-gene, SNV/indel etiology).

Clinical criteria & differential diagnosis. No formal consensus criteria exist (recently delineated). Key differentials among sclerosing/craniotubular dysplasias:

Condition	Gene	Pathway	Distinguishing features
CMDOH	<i>AXIN1</i> (recessive, DIX truncation)	Wnt gain	Sandwich vertebrae + metaphyseal flaring + hip dysplasia + macrocephaly
Craniotubular dysplasia, Ikegawa type (CTDI, OMIM #619727)	<i>TMEM53</i>	BMP-SMAD dysregulation	Skull hyperostosis, childhood blindness (PMID: 33824347, PMID: 39084544, PMID: 41408477)
Sclerosteosis / Van Buchem	<i>SOST</i> / <i>SOST</i> enhancer	Wnt (loss of inhibitor)	High bone mass, syndactyly (sclerosteosis)
Osteopetrosis	<i>CLCN7</i> , <i>TCIRG1</i> , etc.	Osteoclast defect	Fractures, marrow failure, cranial nerve palsies
Endosteal hyperostosis	<i>LRP4/5</i> , <i>SOST</i>	Wnt	Cortical thickening pattern

Screening. In known families, **cascade carrier testing** and, where desired, **prenatal/preimplantation genetic testing** for the familial *AXIN1* variants are options. No population newborn screening exists.

Section 11 — Outcome / Prognosis

- **Survival/mortality:** No systematic survival data. Not reported to be lethal in childhood; the oldest reported individual was 15.5 years. Life expectancy is undefined.
- **Morbidity:** Driven by skeletal complications — potential **cranial-nerve compromise** from skull-base hyperostosis, **orthopedic morbidity** from hip dysplasia, and functional impact of **developmental delay**; cardiac malformations and hematological anomalies contribute variably.
- **Disease course:** **Chronic, progressive** sclerosis. Recovery potential is limited; no disease-modifying therapy is in clinical use.
- **Prognostic factors:** Presumed to include severity of cranial/skull-base involvement (nerve compression risk), cardiac status, and degree of developmental delay. No validated prognostic biomarkers.
- **QoL measures:** Not formally assessed.

Given the small cohort, all prognostic statements carry substantial uncertainty.

Section 12 — Treatment

No disease-specific, approved therapy exists. Management is **supportive and multidisciplinary**.

- **Supportive/rehabilitative:** Orthopedic management of **hip dysplasia** (bracing/surgery as indicated); monitoring for and surgical decompression of **skull-base/foraminal narrowing** if cranial-nerve compromise develops; cardiology management of heart malformations; hematology follow-up; developmental/physical/occupational therapy as needed. NCIT concepts: Supportive Care (NCIT:C15277), Orthopedic Surgery (NCIT:C16536), Physical Therapy (NCIT:C15367).
 - **Pharmacotherapy:** No established drug therapy. No pharmacogenomic guidance.
 - **Advanced/experimental (mechanistically rational, not clinical): Tankyrase inhibition (e.g., XAV939)** is an in-vitro-validated concept — it stabilizes AXIN1/AXIN2 and normalizes Wnt overactivity in mutant cells ([PMID: 37582359](#)). This has **not** been tested in patients; systemic tankyrase inhibition carries known intestinal/skeletal toxicity concerns and would require careful therapeutic-window definition. No gene/cell/RNA therapies are in development for CMDOH.
 - **Treatment strategy:** Individualized, symptom-directed. Any future targeted (Wnt-normalizing) approach would most plausibly target the **early childhood** window of active bone modeling.
-

Section 13 — Prevention

- **Primary prevention:** Not possible for a Mendelian recessive disorder beyond reproductive genetic counseling.
- **Genetic counseling & reproductive options: Central preventive measure.** For carrier couples (25% recurrence risk per pregnancy), options include **carrier/cascade testing, prenatal diagnosis, and preimplantation genetic testing (PGT)** for the familial *AXIN1* variants. Counseling on **consanguinity**-associated recessive risk is relevant.

- **Secondary/tertiary prevention:** Early imaging surveillance to detect and manage skull-base sclerosis (prevent nerve compression), hip surveillance, and cardiac/hematologic monitoring to prevent complications.
- **Immunization / public health / environmental:** Not applicable.

Section 14 — Other Species / Natural Disease

- **Orthologous gene:** *Axin1* is conserved across vertebrates. Mouse *Axin1* (NCBI Gene ID 12005) was originally identified as the **Fused** locus ([PMID: 9230313](#)); "Axin is a novel inhibitor of Wnt signaling." Taxonomy: *Mus musculus* (NCBI:txid10090), *Danio rerio*, etc.
- **Naturally occurring disease:** No naturally occurring CMDOH-equivalent has been catalogued in companion animals or wildlife (no OMIA entry identified for this specific disorder).
- **Comparative biology:** The canonical Wnt/ β -catenin pathway and AXIN scaffold function are **deeply evolutionarily conserved**, underpinning the validity of mouse models. Classic mouse *Fused/Axin1* mutant alleles cause **axial duplication**, reflecting Wnt's role in body-axis patterning.
- **Zoonotic potential / cross-species transmission:** Not applicable (genetic disorder).

Section 15 — Model Organisms

- **Mouse — constitutive knockout:** *Axin1*-null is **embryonic lethal**, precluding study of the adult skeletal phenotype and underscoring why viable human disease requires **hypomorphic** alleles.
- **Mouse — conditional osteoblast-lineage deletion (*Axin1*^{Osx}, *Osterix-Cre*):** Increased β -catenin, elevated **OPG and OPG:RANKL ratio**, decreased TRAP⁺/MMP9⁺/cathepsin-K⁺ **osteoclasts**, suppressed osteoclastogenesis, and delayed postnatal bone growth — directly **recapitulating the human biopsy finding** of increased osteoblast/reduced osteoclast activity (Bone Research 2020, PMC7424530).
- **Mouse — limb-mesenchyme deletion (*Prrx1-Cre*):** Elevated β -catenin producing a fibular-hemimelia/tarsal-coalition phenotype, **partially rescued by deleting one β -catenin allele**

(bone volume 92%→71%), formally demonstrating **β-catenin dependence** (eLife 2023, PMC9815809).

- **Paralog — Axin2 knockout:** Craniosynostosis and increased trabecular bone mass, illustrating overlapping AXIN function in skeletal Wnt regulation and supporting AXIN2 as a compensatory modifier.
- **Cellular models:** Patient-derived primary cells and **CRISPR genome-edited cells** carrying p.Arg723 / p.Asp796Glufs6, plus HEK293T reporter systems, demonstrated enhanced basal Wnt activity and **XAV939 rescue** (PMID: 37582359).
- **Phenotype recapitulation & limitations:** Osteoblast-lineage models faithfully reproduce the **cellular remodeling imbalance** but not the full craniospinal radiographic pattern; constitutive models are lethal. No model reproduces the complete extraskeletal spectrum (cardiac, hematological).
- **Resources:** MGI (mouse), Alliance of Genome Resources, IMPC/IMSR for *Axin1/Axin2* alleles.

Key Findings (with evidence)

F001 — CMDOH is caused by biallelic C-terminal-truncating *AXIN1* variants (autosomal recessive)

Terhal et al. 2023 identified **three homozygous, C-terminally truncating *AXIN1* variants in seven individuals from four families**. Verbatim: *"We identified three homozygous, C-terminally truncating AXIN1 variants in seven individuals from four families affected by macrocephaly, cranial hyperostosis, and vertebral endplate sclerosis. Other frequent findings included hip dysplasia, heart malformations, variable developmental delay, and hematological anomalies."* (PMID: 37582359). This establishes the gene, variant class, inheritance, cohort size, and the core phenotype.

F002 — Mechanism: DIX-domain truncation enhances canonical Wnt/β-catenin signaling; tankyrase inhibition rescues

In patient-derived and genome-edited cells, *"analyses of primary and genome-edited cells harboring the truncating variants revealed enhanced basal canonical Wnt pathway activity."* The truncations impair DIX-domain polymerization but partially retain Wnt-inhibitory function on

overexpression, and *"addition of a tankyrase inhibitor attenuated Wnt overactivity in the AXIN1-mutant model systems"* (PMID: 37582359) — establishing both the causal Wnt-gain mechanism and a candidate targeted therapy.

F003 — AXIN1 is a Wnt-pathway tumor suppressor; complete somatic loss causes cancer, whereas germline hypomorphic truncations cause skeletal dysplasia

AXIN1 is a bona fide tumor suppressor: in hepatocellular carcinoma, *"up to 50% of cases exhibit aberrant activation of the canonical Wnt/ β -catenin pathway, driven by CTNNB1 mutations or inactivating alterations in AXIN1, adenomatous polyposis coli, or ZNRF3, which are mutually exclusive"* (PMID: 42237887). Functional demonstration: *"the WNT pathway and its target gene c-Myc were activated when AXIN1 was missing"* (PMID: 39653061); AXIN1 mutation upregulates CCND1 (p=0.022) with β -catenin–CCND1 correlation $r=0.43$ (PMID: 40344393). The dosage/allele-class contrast explains why germline **hypomorphic** DIX truncations yield a viable Wnt-gain skeletal phenotype rather than malignancy.

F004 — Clinical spectrum and specific variants (OMIM #620558)

Core radiographic features: macrocephaly (+2.2 to +6.1 SD), progressive calvarial/skull-base sclerosis, vertebral endplate ("sandwich") sclerosis, metaphyseal flaring, hypertelorism with low nasal bridge; plus hip dysplasia, heart malformations, variable developmental delay, hematological anomalies, and short stature in a subset. Bone biopsy showed increased osteoblast and reduced osteoclast activity at the growth-plate resorption zone. *"Sclerosing skeletal dysplasias result from an imbalance between bone formation and resorption"* (PMID: 37582359) frames the underlying remodeling defect.

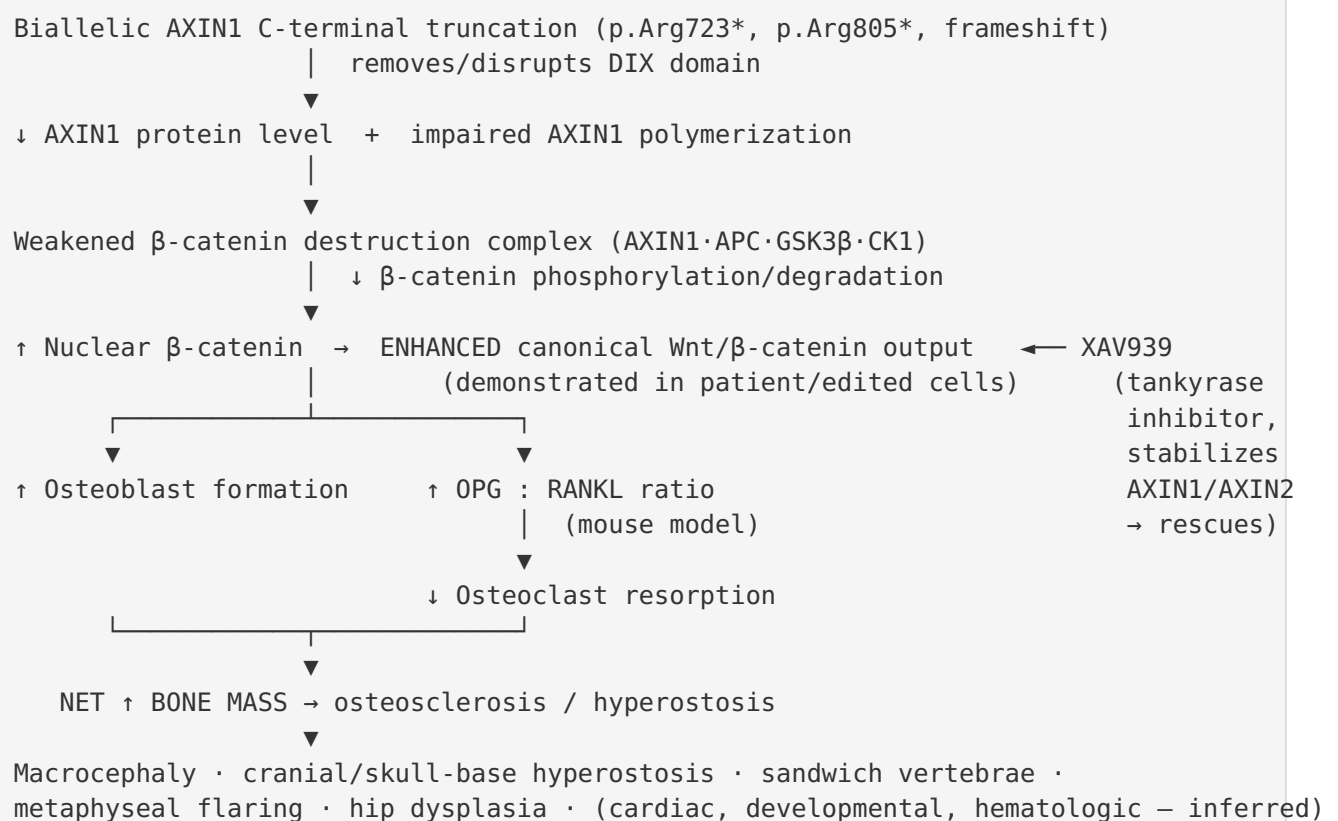
F005 — Mouse *Axin1* models support the Wnt-gain, osteoblast-lineage mechanism

Axin1 was cloned from the mouse *Fused* locus (PMID: 9230313); constitutive knockout is embryonic-lethal. Osteoblast-lineage deletion increases β -catenin and OPG:RANKL and suppresses osteoclastogenesis (mirroring the human biopsy); limb-mesenchyme deletion elevates β -catenin and is partially rescued by removing one β -catenin allele, confirming β -catenin dependence.

F006 — XAV939 dose-dependently rescues Wnt overactivity (partly via AXIN2)

In HEK293T cells engineered with AXIN1 p.Arg723 and p.Asp796Glufs6, XAV939 at 100 nM and 1 μ M produced dose-dependent suppression of Wnt activation; it also rescued AXIN1-knockout cells, implicating **AXIN2 stabilization**. Tankyrases PARylate AXIN1/AXIN2 for degradation, so their inhibition raises AXIN levels and restores destruction-complex activity (PMID: 37582359).

Mechanistic Model (synthesis)



Allele-dosage spectrum of AXIN1 dysfunction:

AXIN1 state	Wnt output	Outcome
Wild-type	Normal (destruction complex intact)	Healthy
Germline biallelic hypomorphic DIX truncation	Moderately ↑ (partial retained function)	CMDOH (viable skeletal dysplasia)
Somatic complete inactivation	Strongly ↑ (full Wnt activation, c-Myc/CCND1)	Cancer (e.g., HCC)

Evidence Base

PMID	Title (abbrev.)	Role in this report
37582359	<i>AXIN1 bi-allelic variants disrupting the C-terminal DIX domain cause CMDOH</i>	Primary defining paper — gene, variants, phenotype, Wnt-gain mechanism, XAV939 rescue
42237887	<i>Wnt/β-catenin in HCC</i>	Establishes AXIN1 as a Wnt tumor suppressor (dosage contrast)
39653061	<i>Loss of AXIN1 & lenvatinib response in HCC</i>	Functional proof loss of AXIN1 activates Wnt/c-Myc
40344393	<i>TP53/AXIN1/CTNNB1/KRAS in Mongolian HCC</i>	AXIN1 mutation ↑ CCND1; β-catenin–CCND1 correlation
9230313	<i>Axin (Fused locus): a novel Wnt inhibitor</i>	Foundational mouse genetics; Axin inhibits Wnt
33824347 , 39084544 , 41408477	<i>TMEM53 craniotubular dysplasia (CTDI)</i>	Key differential — BMP-SMAD (not Wnt) sclerosing dysplasia
18981475	<i>PTH signaling via LRP6/axin</i>	Context: axin/Wnt integration in osteoblast bone formation
26763102	<i>Osteoblast exosomes inhibit Axin1</i>	Context: Axin1 suppression → Wnt activation promotes osteogenesis

Note on evidence strength: The human genotype–phenotype and Wnt-gain findings rest on a **single, well-executed 7-patient study** with strong functional validation. The osteoblast/osteoclast step is supported by **model-organism** data. The tumor-suppressor/dosage framing is **inferential context** from cancer literature, not direct CMDOH data.

Limitations and Knowledge Gaps

1. **Tiny cohort (n=7/4 families):** Frequencies, penetrance, expressivity, and prognosis are imprecise; the phenotype spectrum may broaden as more cases are found.
 2. **No epidemiology:** Prevalence, incidence, carrier frequency, and geographic/ethnic distribution are unquantified.
 3. **Extraskelletal mechanism is inferred:** How Wnt-gain produces the cardiac, hematological, and developmental features is not mechanistically dissected.
 4. **No natural-history/QoL/survival data:** Long-term outcomes and QoL are undefined.
 5. **Therapy is preclinical only:** XAV939 rescue is in-vitro; efficacy, safety, therapeutic window, and delivery in patients are unknown. Systemic tankyrase inhibition has known toxicity.
 6. **No patient omics datasets** (transcriptomic/proteomic/single-cell) exist to refine cell-type-specific mechanisms.
 7. **Modifier genetics unproven:** AXIN2 compensation is inferred from cell rescue, not demonstrated as a clinical modifier.
-

Proposed Follow-up Experiments / Actions

1. **Establish an international CMDOH registry** and GeneMatcher-driven case collection to define prevalence, natural history, penetrance, and expressivity.
2. **Generate a knock-in mouse** carrying a patient-equivalent C-terminal *Axin1* truncation (e.g., R723*-analog) to recapitulate the craniospinal radiographic phenotype and test therapeutic windows.
3. **In-vivo XAV939 / next-generation tankyrase-inhibitor studies** in the knock-in model, defining dose, timing (early bone-modeling window), and skeletal/intestinal toxicity — a prerequisite before any human consideration.

4. **Patient-derived iPSC → osteoblast/osteoclast organoid models** to map cell-type-specific Wnt targets (RNA-seq/single-cell) and OPG:RANKL dynamics directly in human cells.
 5. **Dissect extraskkeletal mechanisms** (cardiac, hematopoietic) via conditional *Axin1*-truncation models in relevant lineages.
 6. **Systematic differential-diagnosis workflow:** incorporate *AXIN1* into sclerosing-dysplasia gene panels and publish radiographic criteria distinguishing CMDOH (Wnt) from TMEM53/CTDI (BMP-SMAD) and osteopetroses.
 7. **Biomarker development:** evaluate serum bone-turnover markers and Wnt-target readouts as potential monitoring/prognostic biomarkers.
-

Report compiled from a 5-iteration autonomous investigation; 6 confirmed findings, 12 papers reviewed. Primary source: Terhal et al., 2023 ([PMID: 37582359](#)).
